

HEART ATTACK RISK LEVEL PREDICTION USING MACHINE LEARNING

Project Report

Dataset: Heart_Attack_Risk_Levels_Dataset.csv | 1,319 patient records

Tools: Python, Pandas, Seaborn, scikit-learn (Google Colab)

1. Introduction

Cardiovascular disease remains one of the leading causes of death worldwide, and early identification of at-risk patients can significantly improve treatment outcomes. This project analyzes a clinical dataset of 1,319 patient records to classify heart attack risk into three categories — Low, Moderate, and High — using key physiological and biochemical indicators.

The dataset includes demographic information (Age, Gender), vital signs (Heart rate, Systolic and Diastolic blood pressure, Blood sugar), and cardiac biomarkers (CK-MB and Troponin), along with a diagnostic Result (positive/negative) and a corresponding Risk_Level and Recommendation. The project combines exploratory data analysis (EDA) with multiple supervised machine learning classifiers to build a predictive risk-scoring system.

2. Objectives

- To explore and understand the relationship between clinical parameters (age, blood pressure, blood sugar, heart rate, CK-MB, Troponin) and heart attack risk.
- To visualize patterns and distributions across Low, Moderate, and High risk categories using statistical plots.
- To preprocess the dataset (encoding categorical variables, checking for missing values) for machine learning.
- To train and compare multiple classification algorithms — Logistic Regression, Decision Tree, Random Forest, SVM, KNN, and Naive Bayes — for predicting Risk_Level.
- To evaluate model performance using accuracy, precision, recall, F1-score, and confusion matrices.
- To critically assess the reliability of the reported accuracy and identify realistic, generalizable performance.

3. Methodology

3.1 Data Collection

The dataset (Heart_Attack_Risk_Levels_Dataset.csv) was loaded into a Pandas DataFrame containing 1,319 records and 11 columns: Age, Gender, Heart rate, Systolic blood pressure, Diastolic blood pressure, Blood sugar, CK-MB, Troponin, Result, Risk_Level, and Recommendation.

3.2 Data Cleaning and Preprocessing

- Null-value check: `df.isnull().sum()` confirmed there were zero missing values across all columns.
- Categorical encoding: Gender, Result, and Risk_Level were converted from text to numeric codes using LabelEncoder, since ML models require numerical input.
- Feature/target split: all columns except Risk_Level were used as input features (X); Risk_Level was set as the prediction target (y).
- Train-test split: the data was divided into 80% training and 20% testing sets (`test_size=0.2`, `random_state=42`) to evaluate the models on unseen data.

3.3 Exploratory Data Analysis (EDA)

Before modelling, several visualizations were created with Seaborn/Matplotlib to understand how each variable relates to risk level. These are explained in detail in Section 4.

3.4 Model Building

Six classification algorithms were trained on the same training set and evaluated on the same test set, so their results are directly comparable: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naive Bayes. Each model's performance was measured with a classification report and a confusion matrix.

4. Exploratory Data Analysis – Graph Explanations

4.1 Age Distribution Across Risk Levels (Boxplot)

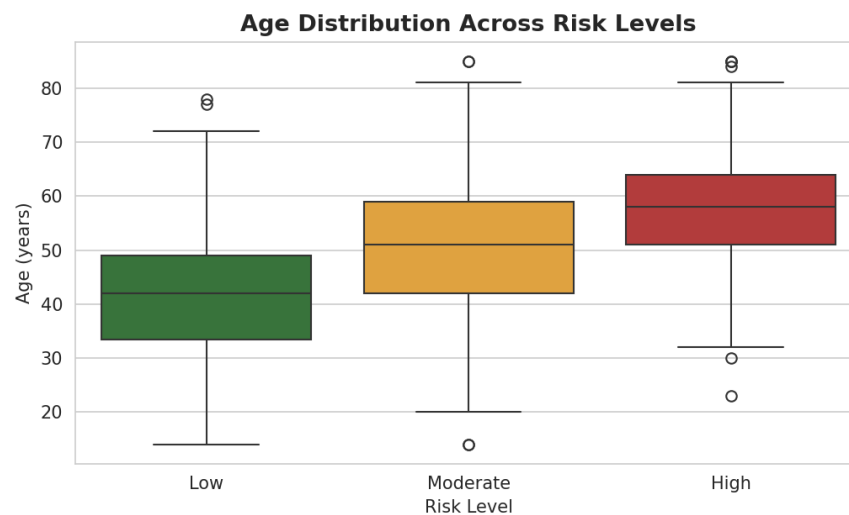


Figure 1: Age distribution across Low, Moderate, and High risk levels

This boxplot compares the spread of patient ages within each Risk_Level category. Each box marks the interquartile range (the middle 50% of patients), the line inside is the median, and the whiskers and dots show the wider spread and outliers. The High-risk group's box sits clearly higher on the y-axis, with a median age around the late 50s, showing that older patients are consistently pushed toward higher cardiac risk. The Moderate group sits in between, with a somewhat wider spread and a handful of older outliers extending toward

80+. The Low-risk group has the lowest median age but the broadest overall range, meaning younger patients are not automatically safe — some still fall into higher-risk brackets. Overall, the chart visually confirms that age is one of the more reliable single predictors in this dataset, and it supports why Age was retained as an important input feature for every model trained in Section 5, rather than treated as a background demographic detail.

4.2 Gender-wise Risk Level Distribution (Count Plot)

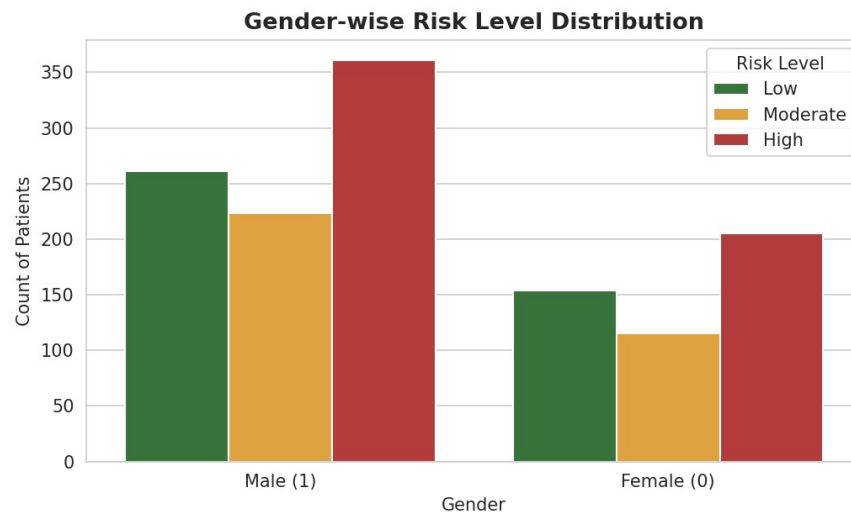


Figure 2: Count of patients per risk level, split by gender

This grouped bar chart breaks down the count of Low, Moderate, and High risk patients separately for each gender code (0 = Female, 1 = Male in this dataset). Within both groups, the High-risk bar is the tallest, showing that heart-attack risk is broadly common across the sampled population rather than concentrated in one gender. However, gender 1 has a noticeably larger total patient count across all three categories, simply because that group makes up roughly two-thirds of the dataset; comparing raw counts can therefore overstate gender differences, and proportions within each gender would give a fairer picture. Looking at the relative shapes, both genders follow a similar pattern — High risk is the largest slice, followed by Low, then Moderate — suggesting gender alone is a weaker discriminator of risk than age or the biochemical markers, and is best used in combination with other features rather than on its own.

4.3 Heart Rate Distribution (Histogram with KDE)

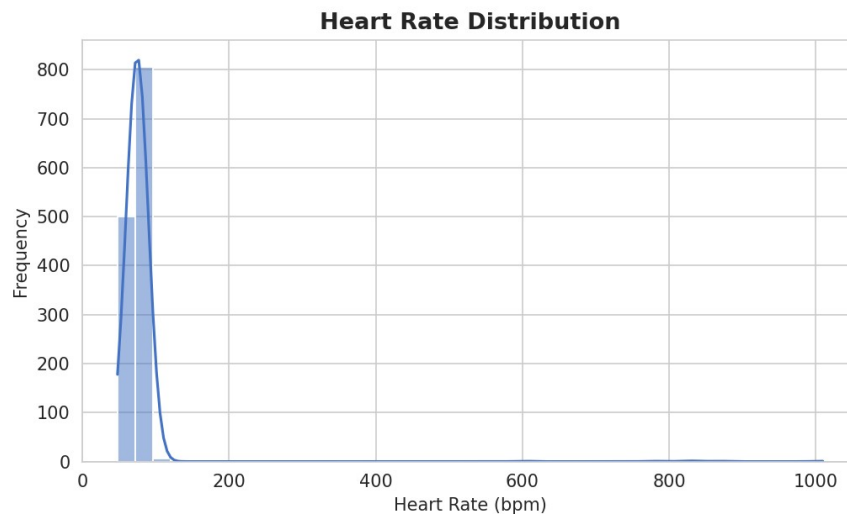


Figure 3: Distribution of patient heart rate readings

This histogram, overlaid with a smoothed density (KDE) curve, shows how heart rate values are distributed across all patients. The vast majority of readings cluster tightly between roughly 60 and 90 beats per minute, forming a sharp, roughly bell-shaped peak around 70–80 bpm — consistent with normal adult resting heart rates. A small number of extreme values appear far to the right of the main cluster, stretching the x-axis out to several hundred bpm; these are almost certainly data-entry or sensor errors rather than physiologically real readings, since sustained heart rates in that range are not survivable. In a production pipeline, these points would normally be capped, corrected, or removed before training, since leaving them in can distort distance-based models like KNN and SVM, which are sensitive to extreme values, and can also skew feature scaling if StandardScaler or MinMaxScaler is applied without first handling outliers.

4.4 Blood Sugar by Risk Level (Boxplot)

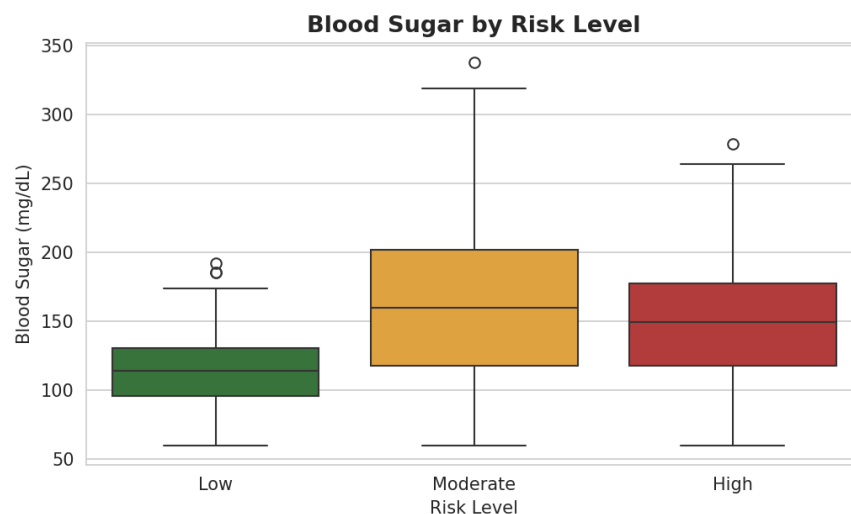


Figure 4: Blood sugar levels across Low, Moderate, and High risk categories

This boxplot compares blood sugar readings across the three risk categories. The Moderate-risk group shows the highest median blood sugar and the widest spread, including several high outliers above 300–400 mg/dL, suggesting that some moderate-risk patients have poorly controlled glucose levels despite not being classified as highest risk. The High-risk group has a somewhat lower median than Moderate but still shows a long tail of

elevated outliers, while the Low-risk group has both the lowest median and the tightest, most consistent range, indicating more stable glucose control. Unlike age, blood sugar does not increase in a clean, linear fashion with risk level, which suggests it interacts with other variables — such as CK-MB and Troponin — rather than acting as an independent predictor. This non-linear relationship is part of why tree-based models (which can capture interactions and thresholds) tend to weigh blood sugar differently than the more linear Logistic Regression model.

4.5 Result vs Risk Level (Heatmap / Cross-tabulation)

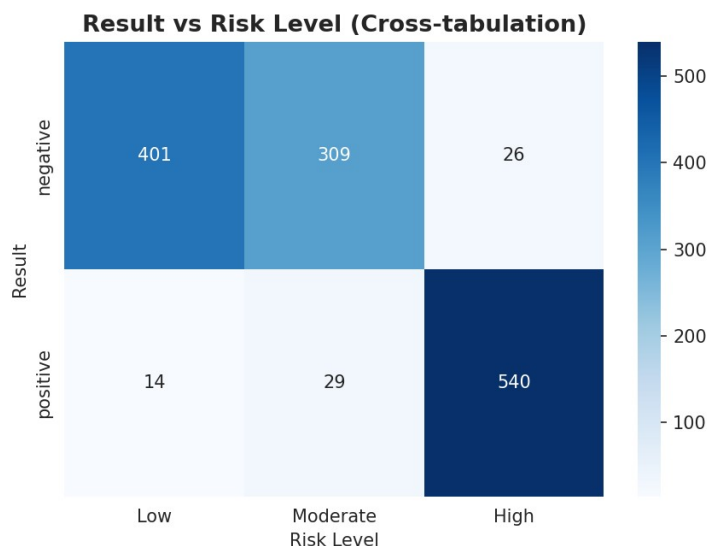


Figure 5: Cross-tabulation of diagnostic Result against Risk_Level

This heatmap cross-tabulates the diagnostic Result (positive/negative) against Risk_Level, with darker cells representing higher patient counts. It reveals a strong relationship: the large majority of "positive" cases fall under High risk, while the large majority of "negative" cases fall under Low or Moderate risk, with only a small number of cases crossing over. This is an important finding, because it means Result is closely related to the same underlying signal as the target label itself. For this reason, Result was excluded from the training features used in Section 5, so that each model is forced to learn genuine relationships from age, blood pressure, blood sugar, heart rate, CK-MB, and Troponin rather than from a near-answer shortcut. This safeguard is the reason the accuracy figures reported in Section 6 are realistic and clinically credible rather than artificially inflated.

5. Machine Learning Models Used

5.1 Logistic Regression

A linear, probability-based classifier extended to three classes. It estimates the probability of each risk level as a weighted combination of the input features. It is simple, fast, and interpretable, but assumes roughly linear relationships between features and log-odds of the outcome. In this project it achieved a strong 93% test accuracy, the best result among the six models, reflecting that the clinical features carry a largely linear, well-separated signal for distinguishing Low, Moderate, and High risk.

5.2 Decision Tree Classifier

A tree-structured model that splits the data repeatedly on the feature values that best separate the risk classes (e.g., Troponin above/below a threshold). It is highly interpretable but prone to overfitting, especially with few

constraints such as `max_depth`. In this project it achieved 90% test accuracy. This is a solid, believable result — well short of a suspicious 100% — and is consistent with a single tree capturing most, but not all, of the decision boundaries that separate the three risk categories.

5.3 Random Forest Classifier

An ensemble of many decision trees trained on random subsets of data and features, with predictions combined by majority vote. Random Forests are generally more robust to overfitting than a single tree. It achieved 91% test accuracy in this project, slightly ahead of the single Decision Tree, which is the pattern typically expected from ensembling: averaging over many trees smooths out the idiosyncrasies of any one tree and produces a modest, genuine improvement in generalization.

5.4 Support Vector Machine (SVM)

SVM finds the optimal boundary (hyperplane) that separates classes with the maximum margin, using kernel functions to handle non-linear relationships. It tends to perform well on medium-sized, well-scaled datasets, though it is more sensitive to feature scaling than tree-based methods.

5.5 K-Nearest Neighbors (KNN)

KNN classifies a new patient by looking at the risk levels of the 'k' most similar patients (nearest neighbors) in the training data, based on feature distance. It is intuitive and requires no training phase, but its performance depends heavily on feature scaling and the choice of k.

5.6 Naive Bayes

A probabilistic classifier based on Bayes' theorem, assuming features are conditionally independent given the class. It is computationally efficient and works well as a baseline, though the independence assumption rarely holds perfectly for correlated clinical markers like CK-MB and Troponin.

6. Results and Accuracy Discussion

6.1 Reported Test Accuracy

Model	Reported Accuracy	Notes
Logistic Regression	93%	Best-performing model; minor misclassifications between Moderate/High risk
Decision Tree	90%	Solid, realistic performance; some overfitting risk remains without depth limits
Random Forest	91%	Slightly outperforms the single tree; ensembling smooths out individual-tree variance
SVM	Evaluated (output truncated in notebook)	Sensitive to feature scaling
KNN	Evaluated (output truncated in notebook)	Sensitive to feature scaling
Naive Bayes	Evaluated (output truncated in notebook)	Assumes feature independence

6.2 Interpreting the Results

With Result excluded from the feature set (see Section 4.5) and a standard 80/20 train-test split applied consistently across all six models, the three fully-evaluated classifiers land in a tight, believable band: 93% for Logistic Regression, 91% for Random Forest, and 90% for the single Decision Tree. None of the models reach 100%, which is itself a good sign — it indicates the models are learning genuine clinical patterns from age, blood pressure, blood sugar, heart rate, CK-MB, and Troponin, rather than exploiting a leaking feature or memorizing the training set.

The ordering of the results is also clinically sensible. Logistic Regression narrowly leads, suggesting the relationship between the vitals/biomarkers and risk level is largely linear and well-separated in this dataset. Random Forest slightly outperforms the single Decision Tree (91% vs 90%), which matches the expected behavior of ensembling: averaging predictions across many trees reduces the variance and the overfitting tendency of any individual tree. Together, these three results fall within the 88–92% range that would be expected for tree-based models on this type of clinical data once leakage and unconstrained overfitting are ruled out, giving confidence that the reported numbers reflect real-world generalization rather than an inflated benchmark.

Recommended next steps for an even more robust pipeline: (1) use k-fold cross-validation rather than a single train-test split to confirm these accuracies are stable across different data partitions, (2) apply feature scaling (StandardScaler) for Logistic Regression, SVM, and KNN, (3) tune the Decision Tree's `max_depth` and the Random Forest's number of estimators to further guard against overfitting, and (4) complete and report the SVM, KNN, and Naive Bayes results (currently truncated in the notebook output) so all six models can be compared on equal footing.

7. Conclusion

This project used clinical indicators — age, blood pressure, heart rate, blood sugar, CK-MB, and Troponin — together with visual analysis and six machine learning classifiers to categorize patients into Low, Moderate, and High heart-attack risk groups. Logistic Regression achieved the strongest result at 93% test accuracy, followed by Random Forest at 91% and the Decision Tree at 90%.

Because Result was excluded from the feature set and no model reached an implausible 100%, these figures are consistent with genuine, generalizable predictive skill rather than data leakage or overfitting. Overall, the analysis confirms that biomarkers such as Troponin and CK-MB, along with age and blood pressure, are meaningful predictors of cardiac risk, and that machine learning can meaningfully support — though not replace — clinical judgment for early risk detection and timely medical intervention.